

DSBDA Oral Questions with Answers

Unit I: Introduction to Data Science and Big Data

Q1. What is Data Science?

Data Science is the process of collecting, cleaning, analyzing, and interpreting data to extract useful knowledge and make decisions.

Q2. What is Big Data?

Big Data means very large, complex, and fast-growing data that cannot be handled easily using traditional database systems.

Q3. Why is Data Science needed?

Data Science is needed to find patterns, predict future outcomes, improve decision-making, and solve real-world problems using data.

Q4. What is data explosion?

Data explosion means the rapid increase in the amount of data generated from sources like social media, sensors, mobile phones, websites, and transactions.

Q5. What are the 5 V's of Big Data?

The 5 V's are:

V	Meaning
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Volume	Huge amount of data
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Velocity	Speed of data generation
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Variety	Different types of data
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Veracity	Accuracy and trustworthiness of data
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Value	Useful information obtained from data
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Q6. What are applications of Data Science?

Applications of Data Science include healthcare, banking, recommendation systems, fraud detection, weather forecasting, education, business analytics, and social media analysis.

Q7. What is the difference between Data Science and Business Intelligence?

Business Intelligence	Data Science
Focuses on past and present data	Focuses on past, present, and future prediction
Mostly uses structured data	Uses structured and unstructured data
Mainly descriptive	Descriptive, predictive, and prescriptive
Uses reports and dashboards	Uses machine learning and statistics

Q8. What is Data Science Life Cycle?

Data Science Life Cycle includes:

1. Data collection
2. Data cleaning
3. Data preparation
4. Data analysis
5. Model building
6. Model evaluation
7. Deployment
8. Communication of results

Q9. What are data types?

Data types describe the nature of data. Common data types are numerical, categorical, text, date-time, Boolean, structured, semi-structured, and unstructured.

Q10. What is data collection?

Data collection is the process of gathering data from different sources such as surveys, files, sensors, databases, APIs, and websites.

Q11. What is data wrangling?

Data wrangling is the process of cleaning, transforming, and preparing raw data into a useful format for analysis.

Q12. Why is data wrangling needed?

Data wrangling is needed because raw data may contain missing values, duplicate values, wrong format, noise, and inconsistent data.

Q13. What is data cleaning?

Data cleaning means removing or correcting incorrect, incomplete, duplicate, or irrelevant data.

Q14. What is data integration?

Data integration means combining data from multiple sources into a single useful dataset.

Q15. What is data reduction?

Data reduction means reducing the size of data while keeping important information.

Q16. What is data transformation?

Data transformation means converting data from one format or scale to another. Example: converting marks into grades or normalizing values.

Q17. What is data discretization?

Data discretization means converting continuous values into categories.

Example: Age can be divided into child, young, adult, and senior.

Unit II: Statistical Inference

Q1. Why is statistics needed in Data Science?

Statistics is needed to understand data, summarize data, find relationships, make predictions, and test assumptions.

Q2. What is mean?

Mean is the average value of data.

Formula:

Mean = Sum of values / Number of values

Q3. What is median?

Median is the middle value of data when values are arranged in ascending or descending order.

Q4. What is mode?

Mode is the value that appears most frequently in the dataset.

Q5. What is mid-range?

Mid-range is the average of the minimum and maximum value.

Formula:

Mid-range = (Minimum + Maximum) / 2

Q6. What is range?

Range is the difference between the maximum and minimum value.

Formula:

Range = Maximum - Minimum

Q7. What is variance?

Variance measures how far data values are spread from the mean.

Q8. What is standard deviation?

Standard deviation shows the spread of data from the mean. A higher standard deviation means data is more spread out.

Q9. What is mean deviation?

Mean deviation is the average distance of each data value from the mean.

Q10. Difference between variance and standard deviation?

Variance	Standard Deviation
Square of deviation from mean	Square root of variance
Unit is squared	Unit is same as original data
Less easy to interpret	Easier to interpret

Q11. What is Bayes theorem?

Bayes theorem is used to calculate conditional probability.

Formula:

$$P(A|B) = P(B|A) \times P(A) / P(B)$$

Q12. What is hypothesis?

A hypothesis is an assumption or statement made about a population, which is tested using sample data.

Q13. What is hypothesis testing?

Hypothesis testing is a statistical method used to accept or reject an assumption based on data.

Q14. What is null hypothesis?

Null hypothesis, written as H_0 , states that there is no significant difference or relationship.

Q15. What is alternative hypothesis?

Alternative hypothesis, written as H_1 , states that there is a significant difference or relationship.

Q16. What is Pearson correlation?

Pearson correlation measures the strength and direction of a linear relationship between two variables.

Q17. What is the range of correlation coefficient?

Correlation coefficient ranges from -1 to +1.

Value Meaning

- +1 Perfect positive correlation
- 1 Perfect negative correlation
- 0 No linear correlation

Q18. What is Chi-Square test?

Chi-Square test is used to check the relationship between categorical variables.

Q19. What is t-test?

A t-test is used to compare the means of two groups.

Q20. Example of t-test?

We can use a t-test to compare average marks of two divisions or average salary of two groups.

Unit III: Big Data Analytics Life Cycle

Q1. What is Big Data Analytics?

Big Data Analytics is the process of analyzing large and complex data to find patterns, trends, and useful information.

Q2. What are sources of Big Data?

Sources include social media, sensors, mobile devices, transaction systems, websites, emails, machines, and log files.

Q3. What is Big Data Analytics Life Cycle?

It is a step-by-step process used to solve big data problems.

Q4. What are the phases of Big Data Analytics Life Cycle?

The phases are:

1. Discovery
2. Data Preparation
3. Model Planning
4. Model Building
5. Communication Results
6. Operationalize

Q5. Explain Discovery phase.

In the discovery phase, the business problem is understood, objectives are defined, and required data sources are identified.

Q6. Explain Data Preparation phase.

In this phase, data is collected, cleaned, transformed, and prepared for analysis.

Q7. Explain Model Planning phase.

In this phase, suitable algorithms, methods, and tools are selected for solving the problem.

Q8. Explain Model Building phase.

In this phase, the actual model is created, trained, and tested using data.

Q9. Explain Communication Results phase.

In this phase, results are presented using reports, graphs, dashboards, and conclusions.

Q10. Explain Operationalize phase.

In this phase, the model is deployed in a real-world environment for actual use.

Q11. What is GINA case study?

GINA stands for Global Innovation Network and Analysis. It is a case study used to analyze innovation networks and relationships using data analytics.

Q12. Why is communication important in analytics?

Communication is important because technical results must be explained clearly to business users or decision makers.

Q13. What is model building?

Model building means creating a machine learning or statistical model using prepared data.

Q14. Why is data preparation important?

Data preparation is important because poor-quality data gives poor results.

Q15. What is the output of the analytics life cycle?

The output is useful insight, prediction, decision support, or a deployed analytical model.

Unit IV: Predictive Big Data Analytics with Python

Q1. What is predictive analytics?

Predictive analytics uses past data to predict future outcomes.

Q2. What is descriptive analytics?

Descriptive analytics explains what happened in the past using reports, charts, and summaries.

Q3. What is prescriptive analytics?

Prescriptive analytics suggests what action should be taken to get the best result.

Q4. Difference between predictive, descriptive, and prescriptive analytics?

Type	Meaning	Example
Descriptive	What happened?	Monthly sales report
Predictive	What may happen?	Future sales prediction
Prescriptive	What should we do?	Best marketing strategy

Q5. What are essential Python libraries for Data Science?

Important libraries are:

- NumPy
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn

Q6. What is NumPy?

NumPy is used for numerical calculations and array operations.

Q7. What is Pandas?

Pandas is used for data manipulation and analysis using DataFrame.

Q8. What is Matplotlib?

Matplotlib is used for creating graphs and charts.

Q9. What is Scikit-learn?

Scikit-learn is a Python library used for machine learning algorithms like regression, classification, clustering, and model evaluation.

Q10. What is data preprocessing?

Data preprocessing means cleaning and preparing raw data before applying machine learning algorithms.

Q11. How do you handle missing data?

Missing data can be handled by:

- Removing rows
- Filling with mean
- Filling with median
- Filling with mode
- Forward fill or backward fill

Q12. Why do we remove duplicate data?

Duplicate data is removed because it can affect accuracy and produce biased results.

Q13. What is transformation using mapping?

Mapping means replacing values using a dictionary or function.

Example: Male = 1, Female = 0.

Q14. What is Apriori algorithm?

Apriori is an association rule mining algorithm used to find frequent itemsets.

Q15. Example of Apriori algorithm?

In a shopping dataset, if customers buy bread and butter together frequently, Apriori can find this relationship.

Q16. What is FP-Growth?

FP-Growth is an efficient algorithm for frequent pattern mining. It is faster than Apriori because it avoids repeated scanning of the database.

Q17. What is regression?

Regression is used to predict continuous numeric values.

Example: House price prediction.

Q18. What is linear regression?

Linear regression predicts a continuous value using a straight-line relationship.

Equation:

$$y = mx + c$$

Q19. What is logistic regression?

Logistic regression is used for classification problems. It gives output in the form of probability between 0 and 1.

Q20. Difference between linear regression and logistic regression?

Linear Regression	Logistic Regression
Used for prediction of continuous values	Used for classification
Output is numeric	Output is class/probability
Example: house price	Example: purchased/not purchased

Q21. What is classification?

Classification is used to predict categorical output.

Example: Spam or not spam, Setosa or Versicolor.

Q22. What is Naïve Bayes?

Naïve Bayes is a classification algorithm based on Bayes theorem.

Q23. Why is Naïve Bayes called naïve?

It is called naïve because it assumes that all features are independent of each other.

Q24. What is decision tree?

A decision tree is a supervised learning algorithm that makes decisions using tree-like conditions.

Q25. What is IRIS dataset?

IRIS dataset contains flower measurements such as sepal length, sepal width, petal length, and petal width. It is used for classification of Iris flowers.

Unit V: Big Data Analytics and Model Evaluation

Q1. What is clustering?

Clustering is an unsupervised learning technique that groups similar data points together.

Q2. What is K-Means clustering?

K-Means is a clustering algorithm that divides data into K groups based on similarity.

Q3. What does K mean in K-Means?

K means the number of clusters.

Q4. What is centroid?

Centroid is the center point of a cluster.

Q5. Steps of K-Means algorithm?

1. Choose number of clusters K
2. Select initial centroids
3. Assign data points to nearest centroid
4. Update centroids
5. Repeat until clusters become stable

Q6. What is hierarchical clustering?

Hierarchical clustering creates a tree-like structure of clusters called a dendrogram.

Q7. What is time-series analysis?

Time-series analysis is used to analyze data collected over time.

Example: Stock price, temperature, sales data.

Q8. What is text analysis?

Text analysis means extracting useful information from text data.

Q9. What is text preprocessing?

Text preprocessing means cleaning text before analysis. It includes tokenization, stop word removal, stemming, and lemmatization.

Q10. What is Bag of Words?

Bag of Words is a method of converting text into numerical form by counting word occurrences.

Q11. What is TF-IDF?

TF-IDF stands for Term Frequency-Inverse Document Frequency. It shows the importance of a word in a document.

Q12. What is social network analysis?

Social network analysis studies relationships between people, groups, or organizations using graphs and networks.

Q13. What is business analysis?

Business analysis means analyzing business data to improve decision-making and performance.

Q14. What is model evaluation?

Model evaluation means checking how well a machine learning model performs.

Q15. What is confusion matrix?

A confusion matrix is a table used to evaluate classification model performance.

Q16. What are TP, TN, FP, FN?**Term Meaning**

TP Actual positive and predicted positive

TN Actual negative and predicted negative

FP Actual negative but predicted positive

FN Actual positive but predicted negative

Q17. What is accuracy?

Accuracy tells how many predictions are correct.

Formula:

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Q18. What is precision?

Precision tells how many predicted positive values are actually positive.

Formula:

$$\text{Precision} = TP / (TP + FP)$$

Q19. What is recall?

Recall tells how many actual positive values are correctly predicted.

Formula:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Q20. What is F1-score?

F1-score is the harmonic mean of precision and recall.

Formula:

$$\text{F1 Score} = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$$

Q21. What is holdout method?

Holdout method divides the dataset into training and testing parts.

Q22. What is random subsampling?

Random subsampling repeatedly splits data randomly into training and testing sets for evaluation.

Q23. What is parameter tuning?

Parameter tuning means selecting the best parameter values to improve model performance.

Q24. What is AUC-ROC curve?

AUC-ROC curve is used to evaluate classification model performance. A higher AUC means a better model.

Q25. What is elbow plot?

Elbow plot is used to find the best value of K in K-Means clustering.

Unit VI: Data Visualization and Hadoop

Q1. What is data visualization?

Data visualization is the graphical representation of data using charts, graphs, and plots.

Q2. Why is data visualization important?

It helps to understand patterns, trends, relationships, and outliers easily.

Q3. What are challenges of Big Data visualization?

Challenges include:

- Large data size
- High speed data
- Different data formats
- Data complexity
- Real-time visualization
- Performance issues

Q4. What are types of data visualization?

Types include:

- Line plot
- Bar chart
- Scatter plot
- Histogram
- Box plot
- Pie chart
- Heatmap
- Density plot

Q5. What is a line plot?

A line plot shows changes or trends over time.

Q6. What is a scatter plot?

A scatter plot shows the relationship between two numeric variables.

Q7. What is histogram?

A histogram shows the distribution of numerical data.

Q8. What is density plot?

A density plot shows the probability distribution of continuous data.

Q9. What is box plot?

A box plot shows data distribution using minimum, Q1, median, Q3, and maximum values.

Q10. What are outliers in box plot?

Outliers are values that lie far away from the normal data range.

Q11. What is Hadoop?

Hadoop is an open-source framework used to store and process large data in a distributed environment.

Q12. What is Hadoop ecosystem?

Hadoop ecosystem is a collection of tools used for big data storage, processing, querying, and analysis.

Examples:

- HDFS
- MapReduce
- YARN
- Hive
- Pig
- HBase
- Sqoop
- Flume

Q13. What is HDFS?

HDFS stands for Hadoop Distributed File System. It stores large files across multiple machines.

Q14. What is MapReduce?

MapReduce is a programming model used to process big data in parallel.

Q15. What are Mapper and Reducer?

Mapper processes input data and creates intermediate key-value pairs.
Reducer combines those values and produces final output.

Q16. What is Pig?

Pig is a high-level platform used to analyze large datasets using Pig Latin language.

Q17. What is Hive?

Hive is a data warehouse tool used to query big data using SQL-like language called HiveQL.

Q18. Difference between Pig and Hive?

Pig	Hive
Uses Pig Latin	Uses HiveQL
More suitable for data flow tasks	More suitable for SQL-like queries
Used by programmers	Used by analysts
Procedural	Declarative

Q19. What is Big Data visualization?

Big Data visualization means representing large and complex data visually to understand patterns and insights.

Q20. Which Python libraries are used for visualization?

Matplotlib, Seaborn, Plotly, and Pandas visualization are commonly used.

Very Important Common Oral Questions**Q1. Difference between Data Science and Big Data?**

Data Science	Big Data
Extracts knowledge from data	Refers to huge volume of data
Uses statistics, ML, visualization	Needs distributed storage and processing
Focus is analysis and prediction	Focus is handling large data
Example: prediction model	Example: social media data storage

Q2. Difference between structured, semi-structured, and unstructured data?

Type	Meaning	Example
Structured	Fixed table format	SQL table
Semi-structured	Partially organized	JSON, XML
Unstructured	No fixed format	Images, videos, emails

Q3. Difference between supervised and unsupervised learning?

Supervised Learning	Unsupervised Learning
Uses labeled data	Uses unlabeled data
Output is known	Output is unknown
Regression and classification	Clustering
Example: house price prediction	Example: customer grouping

Q4. Difference between classification and clustering?

Classification	Clustering
Supervised learning	Unsupervised learning
Uses labeled data	Uses unlabeled data
Predicts predefined class	Finds natural groups
Example: spam/not spam	Example: customer segmentation

Q5. Difference between regression and classification?

Regression

Predicts continuous value

Output is numeric

Example: price prediction

Classification

Predicts category/class

Output is label

Example: disease yes/no

Most Important Formulas

Mean:

Sum of values / Number of values

Range:

Maximum - Minimum

Mid-range:

(Maximum + Minimum) / 2

Variance:

Average of squared deviations from mean

Standard Deviation:

Square root of variance

Bayes Theorem:

$$P(A|B) = P(B|A) \times P(A) / P(B)$$

Accuracy:

$$(TP + TN) / (TP + TN + FP + FN)$$

Precision:

$$TP / (TP + FP)$$

Recall:

$$TP / (TP + FN)$$

F1 Score:

$$2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$$

Top 20 Questions You Must Prepare First

1. What is Data Science?
2. What is Big Data?
3. Explain 5 V's of Big Data.
4. What is data wrangling?
5. Explain data cleaning and data transformation.
6. What is mean, median, and mode?
7. What is standard deviation?
8. What is hypothesis testing?
9. What is Pearson correlation?
10. Explain Big Data Analytics Life Cycle.
11. What is predictive analytics?
12. Difference between regression and classification.
13. Explain Linear Regression.
14. Explain Logistic Regression.
15. What is Naïve Bayes?
16. What is K-Means clustering?
17. What is confusion matrix?
18. Explain accuracy, precision, and recall.
19. What is Hadoop and MapReduce?
20. What is data visualization and why is it important?